第一章

1.6 $\because u = v\cos\theta$

$\cos\theta = \frac{x}{\sqrt{h^2+x^2}}$ $\Rightarrow v = \frac{u\sqrt{h^2+x^2}}{x}$

$a = \frac{dv}{dt} = \frac{u\,d\frac{\sqrt{h^2+x^2}}{x}}{dt} = \frac{u\,d\frac{\sqrt{h^2+x^2}}{x}}{dx}\frac{dx}{dt}$

$= u\,\frac{x(h^2+x^2)^{-\frac{1}{2}} - (h^2+x^2)^{\frac{1}{2}}}{x^2}\,u\,\frac{(h^2+x^2)^{\frac{1}{2}}}{x}$

$= -\frac{h^2u^2}{x^3}$ $\qquad |a| = \frac{h^2u^2}{x^3}$

1.7 (1) $\vec{r} = (5m)\vec{i} + (10m)\vec{j}$ $\qquad |r| = 5\sqrt{5}\,m$

(2) $\vec{v} = [-(10m/s^2)t]\vec{i} + (10m/s)\vec{j}$

$t=1s \Rightarrow \vec{v} = (-10\vec{i} + 10\vec{j})\,m/s$ $\qquad |\vec{v}| = 10\sqrt{2}\,m/s$

(3) $\vec{a} = \frac{d\vec{v}}{dt} = (-10m/s^2)\vec{i}$

$t=1s \Rightarrow \vec{a} = (-10i)\,m/s^2$ $\qquad |\vec{a}| = 10\,m/s^2$

(4)

(4) $V=|\vec{V}|=\sqrt{(100m/s^2)t^2+100m^2/s^2}$ $\Rightarrow a_t=5\sqrt{5}m/s$

$|\vec{a}_t|=\frac{dv}{dt}$

(5) $|\vec{a}_n|=\sqrt{|\vec{a}|^2-|\vec{a}_t|^2}$ $\Rightarrow a_n=5\sqrt{5}m/s$

1.8 $a=\frac{dv}{dt}=-kv^2$

$-\frac{1}{kv^2}dv=dt$

$\int_{v_0}^{v}-\frac{1}{kv^2}dv=\int_0^t dt$

$\Rightarrow \frac{1}{kv}\Big|_{v_0}^{v}=t\Big|_0^t$

$\frac{1}{kv}-\frac{1}{kv_0}=t$

$\Rightarrow v=\frac{v_0}{1+kv_0t}$

$V=\frac{dx}{dt}=\frac{v_0}{1+kv_0t}$

$\Rightarrow dx=\frac{v_0}{1+kv_0t}dt$

$\int_0^x dx=\int_0^t\frac{v_0}{1+kv_0t}dt$

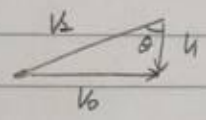
$\Rightarrow x\Big|_0^x=\frac{1}{k}\ln|1+kv_0t|\Big|_0^t$

$\Rightarrow x=\frac{1}{k}\ln(1+kv_0t)$

1.17
$$\begin{cases} v_0=at_0 \\ -v_0t_1+\frac{1}{2}gt_1^2+v_0t_1+\frac{1}{2}at_1^2=h \end{cases} \Rightarrow t_1=0.7s$$

$x=-v_0t_1+\frac{1}{2}gt_1^2$ $\Rightarrow x=0.77m$

1.18



(1) $v_1 = \sqrt{v_2^2 - v_0^2}$ $v_1 = 2\sqrt{21}$ m/s

(2) $\sin\theta = \frac{v_0}{v_2} = \frac{10}{11}$ $\Rightarrow \theta = \arcsin\frac{10}{11}$ $\theta = 65.4°$

1.20 (1) $\vec{x} = (-ut)\vec{i} + (v_0 t - \frac{1}{2}gt^2)\vec{j}$

$\Rightarrow \frac{2v_0 u x + gx^2}{2v_1^2} + y = 0$

$\Rightarrow y = -bx - cx^2$ 其中 $b = 4$, $c = 0.20\,m^{-1}$

(2) $\vec{v} = \frac{d\vec{x}}{dt} = (-u)\vec{i} + (v_0 - gt)\vec{j}$

$\vec{a} = \frac{d\vec{v}}{dt} = -g\vec{j}$

K系与K'系中，小球加速度大小为 $9.8\,m/s^2$，方向竖直向下